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***B.Tech. Degree I Semester Regular/Supplementary & II Semester
Supplementary Examination December 2016***

**GE 15-1103 B CS/IT & GE 15-1203 A CE/EE/ME/SE
ENGINEERING GRAPHICS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

(5 × 12 = 60)

- I. (a) The asymptotes of a hyperbola are at right angles to one another and one point on the curve is 15 mm from each asymptote. Construct the two branches of the curve and draw a tangent and normal at any point on the curve.

OR

- (b) A wire unwinds itself from a drum of 60 mm radius. Draw the locus of the free end of the wire for unwinding through an angle of 180°. Draw a tangent and normal to the curve obtained.

- II. (a) The projectors drawn from HT and VT of a straight line AB are 100 mm apart and those drawn from its ends are 55 mm apart. HT is located 50 mm in front of the VP and VT 110 mm above the HP. The end A of the line lies in the HP. Draw the plan and elevation of the line. Also determine its inclinations to the principal planes of projection and true length of the line.

OR

- (b) A regular pentagonal lamina of 40 mm side rests on the HP with its plane surface vertical and inclined at 30° to VP. Draw its top and front views when one of its sides is perpendicular to HP.

- III. (a) An equilateral triangular pyramid side of base 40 mm and height 70 mm rests on one of its edges of the base on the HP. It is then tilted on this edge such that the base is lifted of the ground by an angle of 45°, keeping the axis of the solid parallel to the VP. Draw its plan and elevation in the new position.

OR

- (b) A right circular cone has 75 m diameter and 100 mm long, rests with its base on the ground. It is cut by a plane inclined at 45° to the HP and perpendicular to the VP and bisecting the axis. Draw the plan and elevation of the cone showing the trace of the section plane on its surface. Also draw the true shape of the section.

(P.T.O.)

- IV. (a) A right circular cone of 60 mm diameter and 90 mm length of the axis rests on its base on the HP. A point P situated on the extreme right end of the base moves around the surface of the cone and finally returns back to the starting point. Find the length of the shortest path the point P will take in covering this distance. Also show the path in plan and elevation.

OR

- (b) A vertical square prism of 45 mm side and 90 mm height rests on its base on the HP with a vertical face making an angle of 45° with VP is intersected by a horizontal square prism of 25 mm side. Their axes intersect with each other. The rectangular face of the horizontal prism is also inclined at 45° to the VP. Draw the projections of the solids showing lines of intersection.

- V. (a) Draw the isometric projection of a hexagonal prism with a hemispherical top touching all the sides. The sides of the hexagonal prism are 45 mm each and height 90 mm.

OR

- (b) Draw the perspective projection of a circle of 75 mm diameter lying on the horizontal plane and whose nearest point is 20 mm behind the picture plane 90 mm to the left of the central plane. Take the height of the eye equal to 70 mm and its distance 65 mm.